



pMVX: Policy-Level Multi-Version Execution for Agentic OS Kernel Self-Tuning

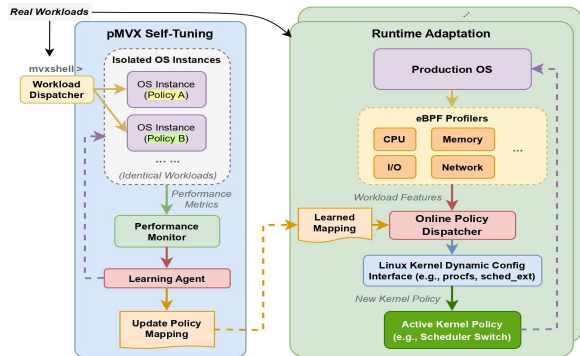
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MOTIVATION

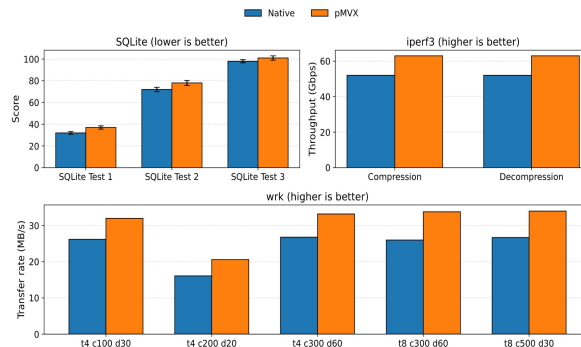
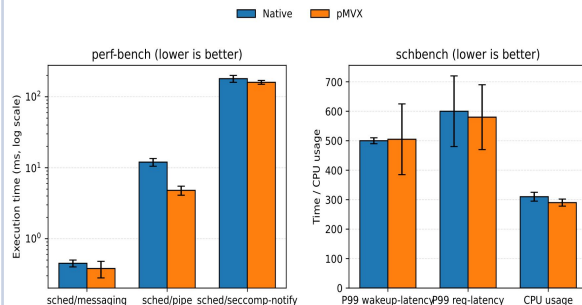
- Kernel policies (e.g., scheduling) are chosen statically, but real workloads are dynamic and heterogeneous.
- Agents managing the OS see only the outcome of the policy they apply never what other policies would have done.
- Production systems can't safely tolerate trial-and-error.

METHODS

- Elevate counterfactual reasoning to different kernel policies.
- pMVX** runs multiple kernel-policy variants concurrently in isolated OS instances.
- Repurposes classic Multi-Version Execution (MVX) which traditionally used for security.
- Store a lightweight policy memory agents can query at runtime.



RESULTS



DISCUSSION

- Prior work (ghost, nest, SchedCP) designs individual schedulers; **pMVX** instead enables comparison among schedulers.
- Architecture is general: applies to memory reclamation, I/O scheduling — any subsystem with multiple policy variants
- Open challenges: cluster-scale evaluation vs. resource cost.

CONCLUSIONS & Future Directions

- pMVX** showed a promising results for kernel scheduling providing an improvement of 20-30% in network heavy workloads.
- In progress work involve integration of various exposed policy variances
- Large scale clusters integration and testing

Further Resources

- Sujot Singh, Eddie Federmeier, Kenan Alghythee, and Xiaoguang Wang. 2026. pMVX: Policy-Level Multi-Version Execution for Agentic OS Kernel Self-Tuning. *In Proceedings of ACM Conference (AgenticOS'26)*. ACM, Pittsburg, PA, USA, 3 pages.

